Session 1: Hello, Python

Question: What is one order line worth?

Concepts

Concept	In daily life	In Excel	The limit
print()	Reading a label out loud	Typing in a cell to see it	Python shows nothing unless you ask
variable	A label on a storage bin	A named cell	Change it, then run the other cells again
int, float, str	Boxes, weights, a product name	Number or text format	"5" is text, not a number
round()	Adding up a receipt	=B2*C2 and ROUND	6 * 2.55 prints 15.299999999999999. Round when you show money
f-string	A fill-in-the-blank label	="Sold "&A2&" units"	Do not forget the f before the quote

Code patterns

```
quantity = 6
unit_price = 2.55
sales_value = quantity * unit_price

print(round(sales_value, 2))

print(f"We sold {quantity} units for {round(sales_value, 2)} pounds")
```

Three common errors

Error	What to do
NameError	A typo in a name. Check the spelling, or run the cells above.
SyntaxError	A bracket or quote is missing. Look at the end of the line.
TypeError	Text plus a number, like "5" + 5. Turn the text into a number with int().

Exercises and expected answers

Exercise	Expected
Predict: What does this cell print? Guess first, then run it.	Guess first, then run it to check
Fill in: The WHITE METAL LANTERN costs 3.39. Its line value, rounded to 2 decimals.	20.34
Predict: Numbers add up. What does text do? Guess both lines first.	Guess first, then run it to check
Fix it: A typo. Fix it so it prints 6.	The cell runs without an error
Fix it: Something is missing. Fix it so it prints hello.	The cell runs without an error
Fix it: Text plus a number. Fix it so it prints 10.	The cell runs without an error
Write it: CREAM CUPID HEARTS COAT HANGER: 8 units at 2.75. Save the line value in hanger_value.	22.0
Write it: One sentence about the first line, with an f-string. Use the variables, not typed numbers.	WHITE HANGING HEART T-LIGHT HOLDER: 6 units = 15.3 pounds
Write it: The price goes up to 2.95. Change unit_price, then compute sales_value again.	17.7
Case: Invoice 536365 starts with the three lines above: the T-light holder (6 at 2.55), the lantern (6 at 3.39) and the coat hanger (8 at 2.75). What are these three lines worth together? Print one sentence.	The first three lines are worth 57.64 pounds
Extra: SET 7 BABUSHKA NESTING BOXES: 2 units at 7.65.	15.3
Extra: Make all three errors on purpose in this cell, one at a time, and read the last line each time.	Your own

The first 12 real order lines (rows 1 to 12 of the data)

Row	Invoice	Description	Qty	Price
1	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	2.55
2	536365	WHITE METAL LANTERN	6	3.39
3	536365	CREAM CUPID HEARTS COAT HANGER	8	2.75
4	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	3.39
5	536365	RED WOOLLY HOTTIE WHITE HEART.	6	3.39
6	536365	SET 7 BABUSHKA NESTING BOXES	2	7.65

7	536365	GLASS STAR FROSTED T-LIGHT HOLDER	6	4.25
8	536366	HAND WARMER UNION JACK	6	1.85
9	536366	HAND WARMER RED POLKA DOT	6	1.85
10	536368	JAM MAKING SET WITH JARS	6	4.25
11	536368	RED COAT RACK PARIS FASHION	3	4.95
12	536368	YELLOW COAT RACK PARIS FASHION	3	4.95